

Case Study 2

HES 3000-002: Sport Performance Analytics

Fall 2026

Overview

Before conducting analysis, it is good to find out what your data actually looks like. In this activity, you will describe one important numeric variable using measures of center, spread, and distribution.

Part 1: Choose what to describe

Identify what one row of your dataset represents (eg: a player, a player-game, a team-game, a training session, etc). Then, choose one numeric variable that matters. Report that variable's units and the number of observations with a usable value.

If you do not have data, use a provided class dataset and complete the same steps. Note which dataset you used.

Part 2: Summarize the variable

Calculate:

- Mean and median
- Minimum, maximum and range
- Standard deviation

Make one plot that shows the distribution

Part 3: Interpretation

Write three or four sentences that address these questions:

- What would you tell someone is a typical value? Would you use the mean or median, and why?
- How much do observations vary? Use at least one measure of spread in your answer.
- What does the distribution show that the summary statistics alone might hide? Consider unusual values, skew, gaps, or clusters.

Bonus:

Compare the variable across a relevant group, such as position, team, season, or competition level.

1. Report the number of observations in each group and compare both center and spread.
2. Make a bar chart to show group means.
3. Describe what you observe. DO NOT make any conclusions or assume causation.